EEP 509: Privacy-Preserving Machine Learning

Course Project Proposal & Plan

Student: Jie Yao

Instructor: Prof. Tamara Bonaci | Electrical & Computer Engineering, UW

Component 1: Project Proposal

When a PII redaction model is deployed in a medical setting it was not trained for, adversaries can craft text that reliably evades detection. Does fine-tuning the model on clinical data close that vulnerability?

This project investigates that question using OpenAI's recently released Privac
y Filter, a 1.5B parameter PII detection model as the target system. We first characterize how distribution shift degrades detection performance on clinical notes, then design adversarial attacks that exploit those degraded failure modes, and finally evaluate whether domain-adaptive fine-tuning on clinical text restores robustness without sacrificing general-domain utility, and if not, where is the Pareto frontier?

1. Problem Description

System. OpenAI Privacy Filter (Apache 2.0, April 2025) is a sparse mixture-of-experts token classifier (1.5B total parameters, 50M active) that detects and masks eight PII categories: person names, addresses, emails, phones, account numbers, URLs, dates, and secrets. It uses a constrained Viterbi decoder over a 128k-token context window. It is designed as a drop-in, low-oversight sanitization layer for high-throughput data pipelines, and is runnable on a single GPU or laptop.

Assets. Patient PII embedded in clinical notes: names, dates of birth, admission dates, addresses, and provider identifiers, regulated under HIPAA. Re-identification from supposedly anonymized records can expose medical history, insurance status, and social context. Critically, downstream systems that consume Privacy Filter output often do so without human review, trusting the tool as a reliable sanitization guarantee; a single missed field can break it entirely.

Attackers. Two profiles: (1) a malicious data recipient: an entity that receives sanitized clinical records and wants to recover PII from them; (2) a malicious data submitter: an insider or external party who crafts input text so that PII survives the redaction pipeline. Both are realistic in federated research collaborations and third-party analytics platforms where sanitized records are shared broadly.

Vulnerabilities. Privacy Filter was trained primarily on general-domain English. Clinical notes differ substantially in vocabulary, syntax, and formatting: names appear in structured fields ("attending: Williams"), dates follow non-standard conventions ("HD#3", "post-op day 2"), and identifiers are embedded in dense medical jargon. The model card explicitly warns of under-detection on out-of-distribution domains. MIMIC-IV-Note, with 330k+ real clinical documents, confirms the severity of this distribution gap.

Threats & Risks. An attacker aware of these blind spots can exploit them via: (1) unicode homoglyph substitution: replacing Latin characters with visually identical characters from other scripts; (2) clinical-syntax obfuscation: embedding PII in structured report formats the model rarely encounters; (3) abbreviation and honorific formatting ("Dr. J.W." vs. full names); (4) whitespace and punctuation injection that disrupts span detection. The consequence is PII exposure in systems that blindly trust the filter, carrying HIPAA liability and direct patient re-identification risk.

2. Proposed Approach

Category: Adversarial ML evaluation + domain-adaptive fine-tuning.

Data strategy. We use two datasets in tandem. AI4Privacy (HuggingFace, no DUA, 2M+ samples) provides labeled PII spans across all eight Privacy Filter categories, the ground-truth needed for quantitative evaluation. MIMIC-IV-Note (PhysioNet, DUA required, 330k+ notes) provides authentic clinical prose: discharge summaries, progress notes, radiology reports. It's for fine-tuning and distribution shift analysis. Since MIMIC is already de-identified, we construct the evaluation set by injecting AI4Privacy PII entities into MIMIC clinical sentences, combining realistic clinical style with known ground-truth labels and giving us more experimental control than any single benchmark.

Four-phase project arc:

- Phase 1 — Baseline audit: run Privacy Filter on the injected clinical evaluation set; measure per-category F1 across all eight PII types; identify which categories degrade most under clinical distribution shift, providing an empirically grounded attack surface map.
- Phase 2 — Adversarial attack: design a targeted evasion suite exploiting the Phase 1 weak spots, such as homoglyph substitution, clinical-syntax obfuscation, abbreviation formatting, whitespace injection; measure evasion rate per attack type and PII category.
- Phase 3 — Fine-tuning defense: LoRA fine-tune Privacy Filter on raw MIMIC-IV-Note prose using UW Hyak HPC resources (e.g., H200, L40S compute nodes); run ablations over training set size and LoRA rank to characterize data efficiency of the defense.
- Phase 4 — Re-evaluation: re-run the full attack suite against the fine-tuned model; compare evasion rates before and after; characterize the residual attack surface.

Expected benefits. A structured vulnerability analysis of a newly released production tool; a clinically fine-tuned Privacy Filter with quantified robustness improvements as a practical artifact; an attack characterization generalizable to other PII detection models facing domain shift. We will also measure whether fine-tuning degrades general-domain PII detection, characterizing the robustness-utility **tradeoff** across LoRA configurations rather than treating fine-tuning as a pure win.

Known drawbacks. MIMIC-IV-Note requires a PhysioNet DUA (~1–2 weeks lead time); AI4Privacy is immediately available for early evaluation work. PII injection introduces some artificiality, entities may not always appear in fully naturalistic clinical positions. LoRA fine-tuning of a 1.5B parameter model requires careful hyperparameter tuning to avoid catastrophic forgetting on general-domain text.

We anticipate that the effectiveness of fine-tuning will vary systematically by attack type. Domain-level attacks such as clinical-syntax obfuscation and abbreviation formatting exploit the model's unfamiliarity with clinical writing style, and we expect fine-tuning on MIMIC-IV-Note prose to meaningfully reduce their evasion rates. Character-level attacks such as unicode homoglyph substitution and whitespace injection operate below the level of language distribution and are orthogonal to domain adaptation; we expect these to remain largely effective even after fine-tuning.

3. References

[1] OpenAI. (2025). OpenAI Privacy Filter Model Card. openai/privacy-filter, Hugging Face.

[2] Johnson, A., Pollard, T., Horng, S., et al. (2023). MIMIC-IV-Note: Deidentified free-text clinical notes. PhysioNet. Version 2.2.

[3] Priva-AI. (2023). AI4Privacy: Large Language Model PII Masking Dataset. HuggingFace Datasets.

[4] Hu, E., Shen, Y., Wallis, P., et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR.

[5] Morris, J., Lifland, E., Yoo, J. Y., et al. (2020). TextAttack: A Framework for Adversarial Attacks in NLP. EMNLP.

[6] Shokri, R., Stronati, M., Song, C., & Shmatikov, V. (2017). Membership Inference Attacks Against Machine Learning Models. IEEE S&P.

Component 2: Project Plan

Final Deliverables

- Technical report (8–10 pages): introduction and motivation, background and related work, threat model, experimental methodology, results across all four phases with honest discussion of limitations, and conclusion.
- Code repository: documented Python codebase covering the PII injection pipeline, baseline evaluation, adversarial attack suite, LoRA fine-tuning scripts, and post-defense evaluation. README with full reproduction instructions targeting UW Hyak.
- Final presentation (8–10 min): threat model walkthrough, key experimental results comparing pre- and post-fine-tuning evasion rates, and a live or recorded demo of adversarial attacks on Privacy Filter.
- Experimental results artifact: saved LoRA model checkpoints, per-experiment result CSVs, and publication-ready figures, F1 curves by PII category, evasion rate heatmaps by attack type, and LoRA ablation plots.

Milestone Plan

Timeline	Milestone
Weeks 1–2	Apply for MIMIC-IV-Note DUA (PhysioNet); set up AI4Privacy dataset immediately; build PII injection pipeline; run Privacy Filter baseline evaluation on injected clinical set; measure per-category F1 across all eight PII types.
Weeks 3–4	Design and implement adversarial attack suite targeting weak PII categories from Phase 1: unicode homoglyph substitution, clinical-syntax obfuscation, abbreviation/honorific formatting, whitespace injection. Measure evasion rates per attack type and PII category.
Weeks 5–6	LoRA fine-tuning of Privacy Filter on MIMIC-IV-Note clinical prose using UW Hyak HPC (e.g., H200, L40S). Run ablations across training set size and LoRA rank. Evaluate on held-out injected clinical test set.
Week 7	Re-run full adversarial attack suite against fine-tuned model. Compare per-category evasion rates before vs. after fine-tuning. Identify and document residual attack surface.
Week 8	Write draft report: introduction, background, threat model, experimental methodology, results, limitations, tradeoffs, and conclusion.
Week 9	Incorporate peer review feedback; finalize and document code repository; prepare final presentation slides and demo.

Responsibilities

This is a solo project. All workstreams, literature review, data pipeline construction, adversarial attack design and implementation, fine-tuning experiments, analysis, writing, and final presentation, are the sole responsibility of the student. The MIMIC-IV-Note DUA application will be submitted during Week 4-5 in parallel with AI4Privacy-based early evaluation, ensuring no blocking dependency on DUA approval for initial work.

Compute & Infrastructure Plan

Phases 1 and 2 (evaluation and adversarial attack) require only CPU or a single consumer GPU and will run locally, leveraging Privacy Filter's 50M active parameter footprint for fast inference. Phase 3 (LoRA fine-tuning) and Phase 4 (post-defense re-evaluation) will use UW Hyak HPC resources. The cluster provides access to high-end GPU nodes (e.g., H200, L40S) well-suited for fine-tuning a 1.5B parameter model with LoRA. Expected fine-tuning time per run is a few hours per configuration, enabling ablations over multiple training set sizes and LoRA ranks within the allocated compute budget. All experiment scripts will be written for reproducibility on Hyak using a shared Conda environment and documented SLURM job scripts included in the code repository.